Exhibit 23 09 05B

Functional Test Procedure Sample

Cal Poly - Vista Grande Dining Facility

System: Airside Systems (AHU, VAVs)
Date: ____________

1. Objective:

a. Confirm satisfactory operation of airside system(s)

2. Equipment:

	EQUIPMENT ID	DESCRIPTION
1.	AHU-_	Air Handler
2.	VAV _ to _	Variable Air Volume (VAV) Terminals
3.		

3. Participants

	NAME	COMPANY	FUNCTION	ROLE
1.				Party filling out this form and witnessing the test
2.				Party operating equipment and executing the test

4. Prerequisite Checklist:

	ITEM	√
1.	Pre-functional checks complete	
2.	TAB complete and zone design lows previously verified	
3.	All safety devices (such as high pressure and low pressure switch) have been checked	
4.	VFD start-up and testing complete	

	ITEM	√
5.	Occupants (if any) notified that testing in progress and zone temperatures will be significantly out of normal range	
6.	No interior dust or fumes in the building	
7.	BAS system able to display real time measured condition in the air handler and in all zones	
8.	All fire dampers open	
9.	Condensate trap at the air handler is primed	

5. Seasonal Testing Note:

Due to the building completion being during winter, this test will be completed in two stages. The first testing will occur during cold weather. The objective of this first stage test is to provide reasonable assurance that the air handler will function properly during lower load conditions. This will prepare the air handler for operation during the beginning of the cooling season. As many of the test procedures as possible will be executed during this first stage of testing, through the use of the methods of artificial loading or partial loading. Tests for full load condition will not be able to be executed until summer.

6. Procedure:

Proced. ID	Sequence/Test	Expected Response	Actual Response	Pass (Y/N)	Note #
1	<u>AHU Disable</u> Standby Check with AHU commanded off by BAS.	Verify by visual inspection that: Supply fan and return fan on AHU slowly ramps down (through VFDs) and comes to a stop Cooling Coil Valve on AHU is Closed OSA Dampers are closed Return Air Damper is open	Supply & Return Fans Off: YES / NO Valve Closed: YES / NO OSA Damper Closed: YES / NO Return Air Damper Open: YES / NO		
2	<u>AHU Enable</u> Operation check with AHU commanded to Occupied Mode by BAS	Verify by visual inspection that: OSA damper opens to min position (economizer verified in a later test) Return Air Damper is open Supply fan and return fan on AHU slowly ramp up (through VFDs) and stabilize to maintain duct static pressure	OSA Damper Open: YES / NO Return Air Damper Open: YES / NO Supply & Return Fans On: YES / NO		

Proced. ID	Sequence/Test	Expected Response	Actual Response	Pass (Y/N)	Note #
3	<u>Basic AHU Operation</u> Lockout any CHW cooling during this test. Set all VAVs to 100% Open and temporarily set duct static pressure setpoint to 0.75”. Observe VFD speed. Slowly raise duct static setpoint until VFD reaches 100% rated speed	VFD speed shall ramp up to maintain static pressure setpoint As duct static pressure setpoint is increased, VFD speed shall ramp up until reaching max speed	Supply Fan Speed (%): _____ Fan speed tracks to maintain static sp: YES / NO Duct static sp to reach max VFD speed: _________ in w.g. Supply Fan Speed (%):_________ Fan speed tracks to maintain static sp: YES / NO Unusual noise or vibration: YES / NO Unusual dust loading in AHU chamber: YES / NO		
4	<u>Zone VAV Box Basic Operation</u> Lockout any CHW cooling during this test. Manually open VAV boxes to 100% open and lock AHU supply fan speed at approximately 80% (or 48 Hz). Observe flows at VAVs.	Flows at each VAV shall be approximately 60 – 80% of max rated VAV flow.	Supply Fan Seed: _______ (%) Duct Static Pressure: ______ in w.g. Air Flows at VAVs normal: YES / NO		

Proced. ID	Sequence/Test	Expected Response	Actual Response	Pass (Y/N)	Note #
4 (Cont.)	Zone VAV Box Basic Operation Fully close one quarter of VAV box dampers and observe the air flow leakage. Note the change in static pressure. Return VAV dampers to their full open position. Repeat procedure for other VAV dampers (one quarter of total VAVs at one time).	Air flow shall be zero (or near zero) with VAV box damper fully closed. Duct static pressure shall increase with the closure of VAV boxes.	Quarter of VAVs closed: (1 of 4) Static pressure: _________ in. w.g. Quarter of VAVs closed: (2 of 4) Static pressure: _________ in. w.g. Quarter of VAVs closed: (3 of 4) Static pressure: _________ in. w.g. Quarter of VAVs closed: (4 of 4) Static pressure: _________ in. w.g. Record individual VAV flows with damper fully closed in VAV detail sheet.		
5	Zone Reheat Coil Basic Operation Lockout any CHW cooling during this test. Open up all boxes to min. flow position. With all reheat coils initially off, observe discharge air temperatures (DAT) for each zone. Open all reheat coils to 100% and observe the increase in supply air temperature. Fully close all reheat coils and observe the decrease in supply air temperature.	Discharge air temperature (DAT) at VAV shall be similar to AHU discharge air temperature (DAT). Discharge air temperature shall increase with opening of reheat coil valves Discharge air temperature shall decrease to normal levels (i.e., close to AHU discharge temp) with closure of reheat coil valves. If discharge air temp does not drop, it can be a sign of leaking HW valves	AHU DAT: ___________ deg.F Range of VAV DATs: ___________ deg.F AHU DAT: ___________ deg.F Range of VAV DATs: ___________ deg.F Discharge air temperatures normal after closure of reheat valves: YES / NO		

Proced. ID	Sequence/Test	Expected Response	Actual Response	Pass (Y/N)	Note #
6	Cooling Coil Valve Control Lock out economizer during this test. Set all VAVs to minimum flow to simulate a low flow condition. Set supply air temp. (SAT) setpoint to 60 deg.F and observe CHW coil response. Lower supply air temperature (SAT) setpoint to 55 deg.F and observe CHW coil response. Open VAVs to achieve max rated airflow. Observe CHW coil response.	Chilled water valve shall modulate to maintain the supply air temperature setpoint. CHW valve shall open up more to maintain the lower supply air temperature setpoint. CHW valve shall open up more to maintain the higher air flow rates.	Supply Fan CFM: _________ MAT / SAT: _______ / ______ deg.F CHW Valve Position: ______ % Open CHW Temp (Sup/Ret): ___/___ deg.F Supply Fan CFM: _________ MAT / SAT: _______ / ______ deg.F CHW Valve Position: ______ % Open CHW Temp (Sup/Ret): ___/___ deg.F Supply Fan CFM: _________ MAT / SAT: _______ / ______ deg.F CHW Valve Position: ______ % Open CHW Temp (Sup/Ret): ___/___ deg.F CHW modulates to achieve setpoint within reasonable time: YES / NO		
7	Minimum Outdoor Air Control Lock out economizer during this test. Set all VAVs to minimum flow to simulate a minimum supply flow condition. Observe outside air (OSA) damper and flow rate. Set all VAVs to max flow to simulate a max design supply flow condition. Observe outside air (OSA) damper and flow rate.	OSA damper shall modulate to maintain the OSA flow CFM setpoint of _______. Damper shall stabilize within 15 minutes. OSA damper shall modulate to maintain the OSA flow CFM setpoint of _______. Damper shall stabilize within 15 minutes.	Supply Fan Speed: _______(%) Supply Fan CFM: _________ OSA CFM: ___________ OSA Damper Position: ________ (%) Time for damper to stabilize: ______ min Supply Fan Speed: _______(%) Supply Fan CFM: _________ OSA CFM: ___________ OSA Damper Position: ________ (%) Time for damper to stabilize: ______ min		

<table>
<tr><th>Proced. ID</th><th>Sequence/Test</th><th>Expected Response</th><th>Actual Response</th><th>Pass (Y/N)</th><th>Note #</th></tr>
<tr><td>8</td><td><u>Economizer Control</u>

Warm OSA Condition (i.e., Economizer Off): Set the system to cooling mode. Simulate a condition such that OSA temp is more than 1 deg.F greater than return air temp, and/or OSA is more than econ high limit temp (i.e., 69 deg.F.)

Cool OSA Condition (i.e., Economizer On): Set the system to cooling mode. Simulate a condition such that OSA temp is less than return air temp, and OSA is less than econ high limit temp (i.e., 69 deg.F.)

Cold OSA Condition (i.e., Economizer On): Set the system to cooling mode. Simulate a condition such that OSA temp is less than supply air temp setpoint.

Ensure building pressures during economizer operation are adequate.</td><td>

OSA damper (OAD) shall remain in min position while CHW cooling coil opens to maintain supply air setpoint. Return air damper (RAD) shall remain fully open.

OAD shall open to 100%. CHW valve shall modulate to maintain supply air setpoint.

OAD shall modulate in association with the RAD to produce air at the supply air temperature setpoint. CHW valve shall remain closed.

Exhaust air damper (EAD) shall modulate to maintain building dp setpoint of 0.02" W.C.</td><td>

OAD remains in min. position:
YES / NO
RAD remains fully open:
YES / NO

OAD damper modulates 100% Open:
YES / NO
RAD modulates fully closed:
YES / NO
CHW valve modulates open to maintain SAT setpoint: YES / NO

OAD and RAD modulate to maintain SAT setpoint: YES / NO
CHW valve remains closed: YES / NO

EAD modulates to maintain building pressure setpoint: YES / NO</td><td></td><td></td></tr>
<tr><td>9</td><td><u>SAT Reset</u>

Supply air temperature (SAT) shall be reset based on OSA temperature. Simulate various OAT conditions and verify SAT resets accordingly.</td><td><table><tr><td>SAT (deg.F)</td><td>OAT (deg.F)</td></tr><tr><td>58</td><td>55 and below</td></tr><tr><td>57</td><td>60</td></tr><tr><td>56</td><td>65</td></tr><tr><td>55</td><td>70 and above</td></tr></table></td><td>SAT setpoint is reset per the provided schedule:
YES / NO</td><td></td><td></td></tr>
</table>

Proced. ID	Sequence/Test	Expected Response	Actual Response	Pass (Y/N)	Note #
10	<u>VFD Response to Zone Air flow Demand</u> Lock out the economizer. Lockout static pressure reset. Set static setpoint to 1.2 in. w.g. Drive 100% of the VAV boxes served by the AHU to minimum flow position.	VFD shall ramp up fan speed to provide more air flow as more VAV box dampers open	*100% of VAVs at Min Flow:* Supply Static SP (in. wg): ______ Supply Fan CFM: ______ Supply Fan Speed: ______ (%) Return Fan CFM: ______ Return Fan Speed: ______ (%) OSA CFM: ______ OSA Damper: ______ (%)		
	Drive 50% of the VAV boxes served by the AHU to minimum flow position.	VFD shall ramp up fan speed to provide more air flow as more VAV box dampers open	*50% of VAVs at Min Flow:* Supply Static SP (in. wg): ______ Supply Fan CFM: ______ Supply Fan Speed: ______ (%) Return Fan CFM: ______ Return Fan Speed: ______ (%) OSA CFM: ______ OSA Damper: ______ (%)		
	Drive 0% of the VAV boxes served by the AHU to minimum flow position (i.e., max flow).	VFD shall ramp up fan speed to provide more air flow as more VAV box dampers open	*0% of VAVs at Min Flow:* Supply Static SP (in. wg): ______ Supply Fan CFM: ______ Supply Fan Speed: ______ (%) Return Fan CFM: ______ Return Fan Speed: ______ (%) OSA CFM: ______ OSA Damper: ______ (%) VFD speed modulates to maintain the required flow at the zone: YES / NO		

Proced. ID	Sequence/Test	Expected Response	Actual Response	Pass (Y/N)	Note #
11	Static Reset Supply air static set-point shall be reset up or down based on pressure requests.	Static set-point shall be increased if there are more than __two__ pressure requests (i.e., VAV damper > 80% open = 1 request; VAV damper > 95% open = 2 requests) Static set-point shall be increased if there are no pressure requests Static set-point High Limit: __1.2_ " w.c. (as determined by air balancer per specified procedure) Static set-point Low Limit: __0.4__ " w.c.	Static set-point resets up satisfactorily: YES / NO Static set-point resets down satisfactorily: YES / NO		
12	Alarms Alarm shall be triggered if the filter pressure drop exceeds 0.8 in. w.c. at design supply air CFM. The alarm shall vary with fan speed as is outlined in the sequence of operations.		Filter Alarm is functional: YES / NO		

Notes: